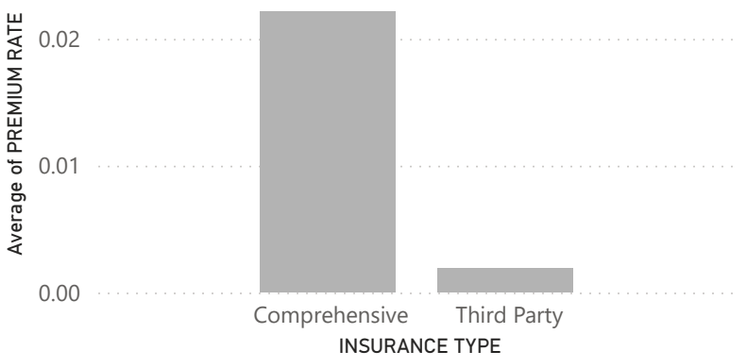
INSURANCE TYPE

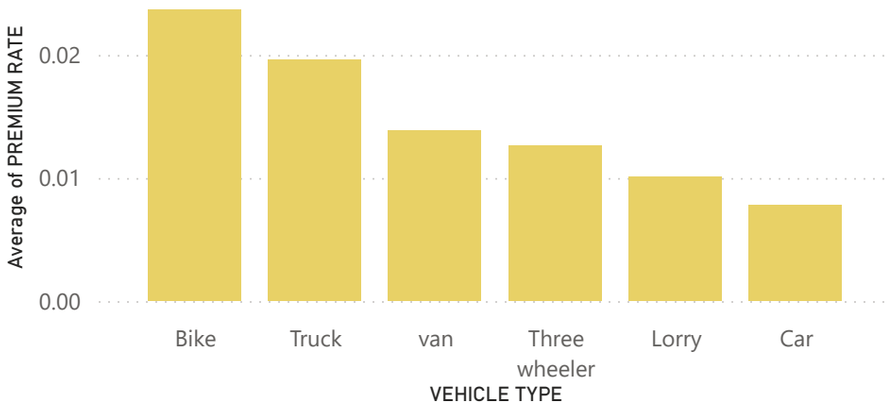
- [ ] Comprehensive
- [ ] Third Party

3998	394	52.031K	1.1995%	7.0022M
Total Policies	Count of VEHICLE MODEL	Average of POLICY PREMIUM	Average of PREMIUM RATE	Average of SUM INSURED

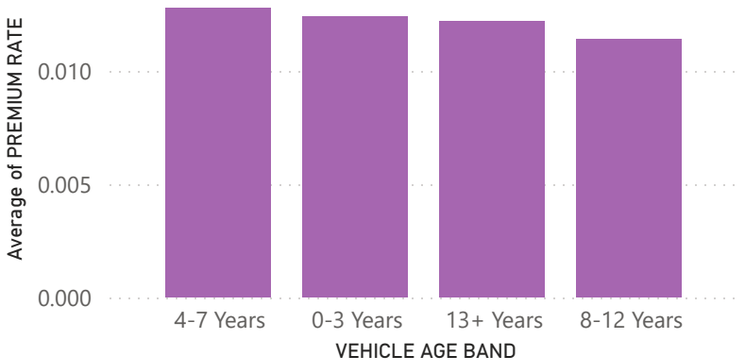
Average of PREMIUM RATE by INSURANCE TYPE



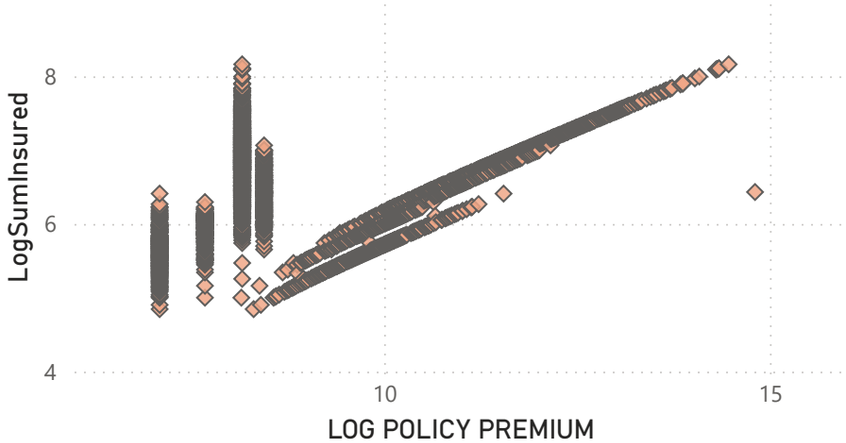
Average of PREMIUM RATE by VEHICLE TYPE



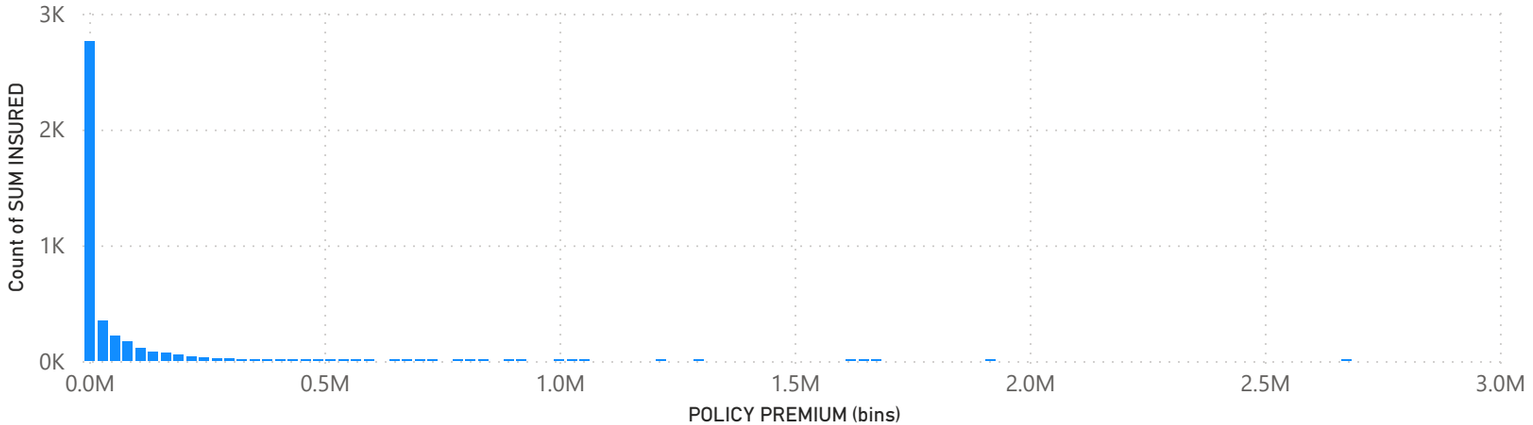
Average of PREMIUM RATE by VEHICLE AGE BAND



Log Policy Premium vs Log Sum Insured

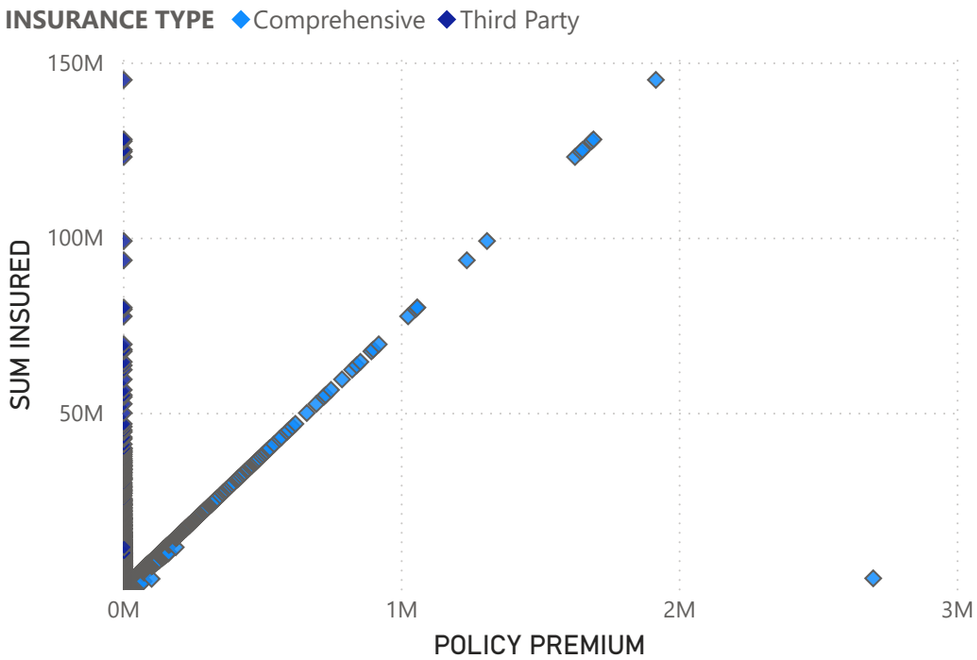


Premium Distribution



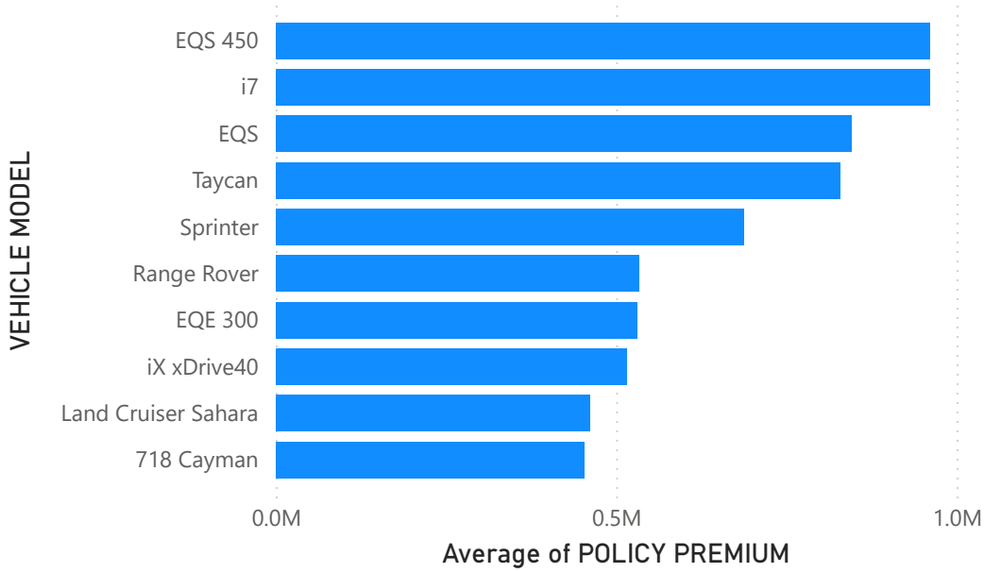
The motor insurance portfolio consists of 3,998 policies with an average premium of approximately 52,000 and an average premium rate of around 1.2%. Premium levels are primarily driven by insurance coverage type and insured vehicle value. Comprehensive policies and cars generate the largest premium volumes. Premium distribution is heavily right-skewed, indicating that a small number of high-value policies contribute disproportionately to total premiums.

Premium vs Sum Insured



Premium Contribution by Segment

Average of POLICY PREMIUM by VEHICLE MODEL



INSURANCE TYPE

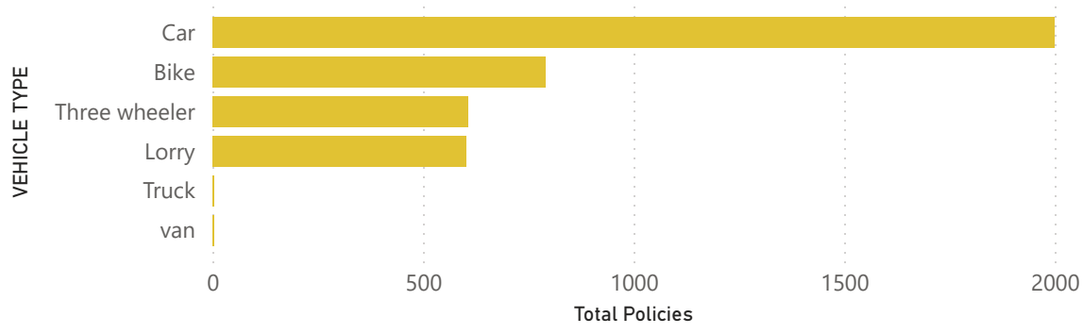
- ☐ Comprehensive
- ☐ Third Party

VEHICLE TYPE	Comprehensive	Third Party
Bike	19,994.30	1,200.00
Car	171,053.14	3,500.00
Lorry	57,413.04	4,650.00
Three wheeler	20,168.27	2,160.00
Truck	35,278.05	
van	63,530.71	

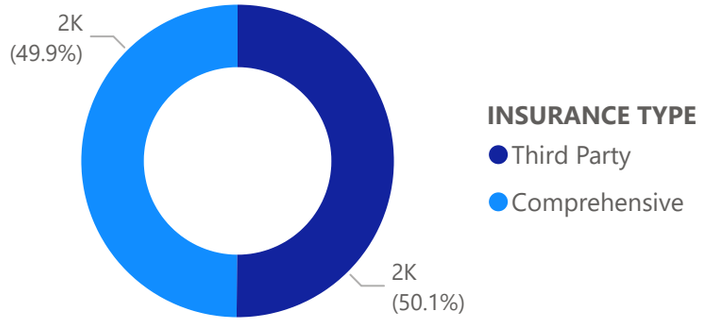
Premium pricing in the portfolio is primarily driven by insured vehicle value and insurance coverage type. The strong linear relationship between policy premium and sum insured suggests a tariff-based pricing structure where premiums scale proportionally with vehicle value. Comprehensive car policies contribute the largest share of premium revenue due to their higher coverage levels and higher insured values. Luxury vehicle models such as EQS, i7, and Taycan exhibit the highest average premiums, reflecting their elevated vehicle values and potential repair costs. Overall, the portfolio's premium concentration is heavily influenced by high-value vehicles insured under comprehensive coverage.

Total Policies by VEHICLE MAKE

Total Policies by VEHICLE TYPE

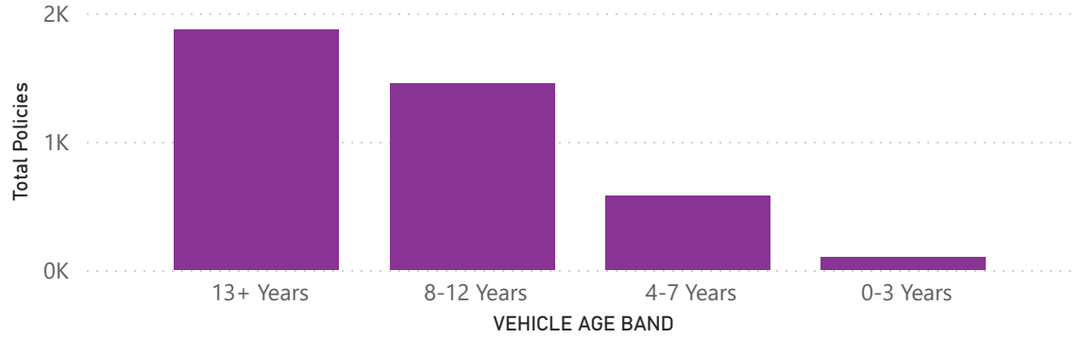


Total Policies by INSURANCE TYPE



Portfolio exposure analysis shows that car policies represent the largest share of the insured portfolio, followed by bikes and three-wheelers. Vehicle manufacturer exposure is concentrated among several major brands, with Bajaj, Toyota, and Honda accounting for a significant portion of total policies. Additionally, most insured vehicles fall within older age bands (8–13+ years), indicating that the portfolio includes a substantial number of aging vehicles.

Total Policies by VEHICLE AGE BAND

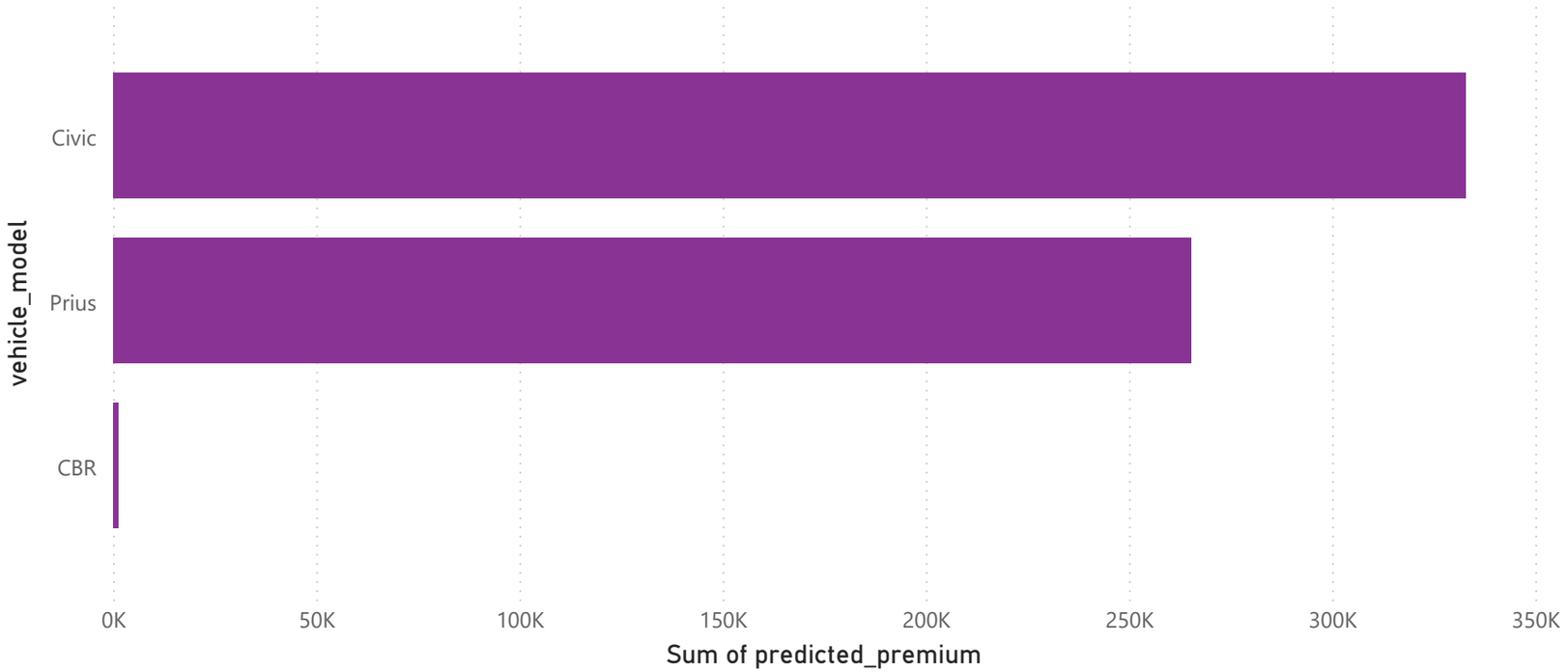


199.74K
Average of predicted_premium

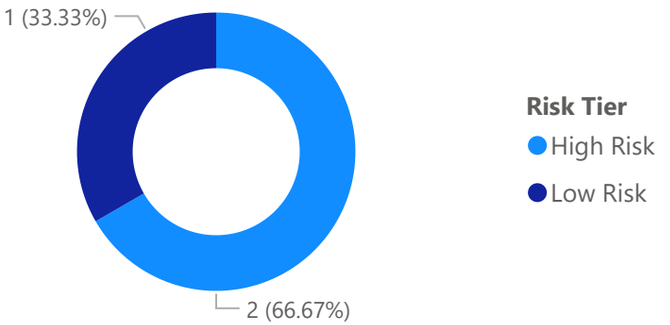
0.8958%
Average of premium_rate

332.82K
Max of predicted_premium

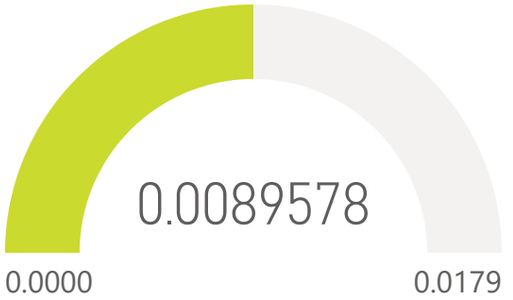
Scenario Comparison Chart



Risk Tier Classification



This page demonstrates how the trained machine learning pricing model can be applied to estimate premiums for new insurance policy scenarios. The simulator generates predicted premiums and premium rates based on vehicle characteristics and insurance coverage, enabling comparison across different policy configurations. By benchmarking simulated premium rates against the portfolio average, policies can be categorized into risk tiers, helping identify relatively lower- or higher-priced scenarios. This illustrates how the model can support underwriting decisions and premium evaluation in practical pricing applications.

insurance_type	vehicle_type	vehicle_make	vehicle_model	Sum of vehicle_age	Sum of sum_insured	Sum of predicted_premium	Sum of premium_rate
Comprehensive	Car	Honda	Civic	1	25000000	332,822.29	0.01
Comprehensive	Car	Toyota	Prius	2	20000000	265,212.15	0.01
Third Party	Bike	Honda	CBR	5	4000000	1,200.00	0.00
Total				**8**	**49000000**	**599,234.44**	**0.03**

Premium Rate Benchmark